fairy: an overview of all functions

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2026-10-24

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fairy turns verbal hypotheses about proportions into order-constrained models and answers the questions the paper "Wait, what are you saying, exactly?" asks of them: What exactly does the hypothesis say (constraints)? Which extreme cases does it allow? How parsimonious is it? How does it relate to other models? This page introduces the idea and lists every function. The vignette `vignette("cringe")` then reproduces the paper's examples.

1 The idea

A hypothesis about `n` proportions `p1, ..., pn` is a set of linear (in)equalities. Together they describe a convex polytope inside the unit cube `[0, 1]^n`. The package describes that polytope in two equivalent ways:

- the **H-representation**: the shortest list of constraints (equalities and facet-defining inequalities);
- the **V-representation**: the extreme cases (vertices) of the model.

Its **volume** is the share of all conceivable proportions the model allows; its inverse is the largest Bayes factor the model can reach against the unconstrained model.

2 Writing a model

A model is one character string. Constraints are separated by `;`.

Write	Meaning
`p1 > p2`	`p1` at least `p2` (`>=` and the unicode symbol mean the same: models are closed sets)
`p1 > p2 > p3`	chain: `p1 > p2` and `p2 > p3`
`p1 = p2`	equality (the model then has volume zero)
`p1 > .3; p1 < .7`	bounds
`{p1, p2} > {p3, p4}`	every `p` on the left above every `p` on the right
`(p1 + p2)/2 > p3`	averages and other linear expressions

`approx_tol = 0.05` (in `fairy_h()`) turns every equality into "within 0.05".

3 Constraints and extreme cases

```
h <- fairy_h("p1 > p2 > p3", n_p = 3)     # H-representation (rcdd format)
fairy_constraints(h)                       # the same, as readable text
#> [1] "- p1 + p2 <= 0" "- p2 + p3 <= 0" "p1 <= 1"        "- p3 <= 0"
fairy_vertices(h)                          # extreme cases, numbers
#>      p_{1} p_{2} p_{3}
#> [1,]     1     1     1
#> [2,]     1     1     0
#> [3,]     1     0     0
#> [4,]     0     0     0
fairy_words(h, labels = c("A", "B", "C"))  # extreme cases in words: "all" / "none"
#>       A    B    C
#> V1 none none none
#> V2  all none none
#> V3  all  all none
#> V4  all  all  all
```

`fairy_v(h)` returns the vertices in rcdd's own format. `fairy_dim()` gives the dimension of the model. Models with an equality have a dimension below `n` and volume zero:

```
c(fairy_dim("p1 > p2 > p3"), fairy_dim("p1 = p2; p2 > p3"))
#> [1] 3 2
```

4 Parsimony

```
fairy_volume("p1 > p2 > p3")$volume          # 1/6
#> [1] 0.1666667
fairy_max_bf("p1 > p2 > p3")                 # 6
#> [1] 6
fairy_info("p1 > p2 > p3")                   # dimension, vertices, constraints, volume, max. BF in one row
#>   dim vertices inequalities equalities    volume max_bf
#> 1   3        4           4          0 0.1666667      6
```

The volume is exact up to about nine dimensions. Above, `fairy_volume()` uses the volesti package (random-walk estimate) if installed and otherwise Monte Carlo (`method = "mc"`). `fairy_parsimony()` is the older name of the same computation.

5 Several models at once

`fairy()` computes constraints, extreme cases, volume and maximal Bayes factor for a named set of models:

```
fairy(c(H1 = "p1 > p2 > p3", H2 = "p1 > p2; p3 > p4"), n_p = 4)
#> $H1
#> fairy model: H1
#>   Model:        p1 > p2 > p3
#>   Parameters:   4
#>   H-rep rows:   6
#>   Vertices:     8
#>   Volume:       0.166667
```

```
#>   Max. BF:       6
#>
#> $H2
#> fairy model: H2
#>   Model:        p1 > p2; p3 > p4
#>   Parameters:   4
#>   H-rep rows:   6
#>   Vertices:     9
#>   Volume:       0.25
#>   Max. BF:      4
```

6 Overlap, nesting, mixtures, comparison

```
a <- "p1 > p2 > p3"; b <- "p1 > p2"
fairy_nested(a, b, n_p = 3)                  # is a nested in b?  TRUE
#> [1] TRUE
ab <- fairy_intersect(a, b, n_p = 3)         # constraints of both hold
mx <- fairy_mixture("p1 > .8; p2 < .2", "p1 < .2; p2 > .8", n_p = 2)   # convex hull of two models
fairy_constraints(mx)
#> [1] "- p2 <= 0"                  "- 5/4 * p1 - 5/4 * p2 <= -1"
#> [3] "- p1 <= 0"                  "p2 <= 1"
#> [5] "p1 + p2 <= 6/5"             "p1 <= 1"
fairy_compare(list(A = a, B = b, C = "p3 > p1"), n_p = 3)        # volumes, overlaps, nesting, max. BF
#>       A        B    C
#> A     0.17     0.17 0
#> B     superset 0.5  0.17
#> C     1D       3D   0.5
#> maxBF    6        2    2
```

`fairy_compare()` returns the table layout of the paper: volumes on the diagonal, overlap volumes above it, below it "subset" / "superset" (printed as the set symbols) or the dimension of the overlap, and the largest Bayes factors in the last row.

7 Replication across labs

```
h2 <- fairy_h("p1 > p2", n_p = 2)
fairy_info(fairy_replicate(h2, r = 3, type = "joint"))
#>   dim vertices inequalities equalities volume max_bf
#> 1   6       27           9          0  0.125      8
```

`type = "identical"` forces all labs to have the same probabilities, `"joint"` lets each lab satisfy the model independently, `"substitutable"` requires every constraint to hold whichever lab's probability is plugged in for each term.

8 Export

`fairy_write_qtest()` and `fairy_write_multinomineq()` write the models as input files for QTest and for the R package multinomineq (one `.tar` archive with one file per model).

9 All functions

Function	What it does
`fairy()`	constraints, extreme cases, volume and max. Bayes factor for one or several models
`fairy_h()`	non-redundant H-representation of a model string
`fairy_v()`	V-representation in rcdd format
`fairy_constraints()`	H-representation as readable text lines
`fairy_vertices()`	extreme cases as a numeric matrix
`fairy_words()`	extreme cases in words ("all" / "none" / fractions)

Function	What it does
`fairy_dim()`	dimension of a model
`fairy_volume()`	parsimony: volume in the unit cube (exact, volesti, or Monte Carlo)
`fairy_max_bf()`	largest possible Bayes factor against the unconstrained model (1 / volume)
`fairy_info()`	dimension, vertices, constraints, volume, max. Bayes factor in one row
`fairy_parsimony()`	older interface to the volume
`fairy_intersect()`	intersection of models
`fairy_mixture()`	mixture model (convex hull of the components)
`fairy_nested()`	is one model nested in another?
`fairy_compare()`	comparison table of a set of models (Table 3 of the paper)
`fairy_replicate()`	replication across labs: identical, joint, substitutable
`fairy_write_qtest()`	write input files for QTest
`fairy_write_multinomineq()`	write input files for multinomineq

Help for any function: `?fairy_volume`; all of them: `help(package = "fairy")`.